

A Hybrid Machine Learning Framework for Accurate Healthcare Disease Prediction

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ABSTRACT

Healthcare data is increasing rapidly, creating significant opportunities for accurate and early disease prediction using machine learning techniques. However, single-model approaches often fail to deliver consistent and high predictive performance across diverse medical datasets. This study proposes a hybrid machine learning framework that integrates feature optimization and ensemble-based classification methods to enhance disease prediction accuracy and robustness. The proposed model is evaluated using healthcare datasets and assessed through performance metrics such as accuracy, precision, recall, and F1-score. Experimental results demonstrate that the hybrid approach outperforms traditional standalone models and offers improved stability in complex clinical data environments. This research contributes to the advancement of intelligent, data-driven healthcare systems for early diagnosis and improved patient outcomes.

KEYWORDS

Hybrid Machine Learning, Healthcare Disease Prediction, Ensemble Learning, Feature Optimization, Medical Data Analysis, Decision Support Systems.